Journal of Crystal Growth 562 (2021) 126088

Contents lists available at ScienceDirect

Journal of Crystal Growth

journal homepage: www.elsevier.com/locate/jcrysgro

Single crystal growth of GaSe via bismuth flux method and its air-stability

Weiwei Chu [a,b], Jiyong Yang [a,c], Liang Li [a,d], Xiangde Zhu [a,*], Mingliang Tian [a,e,*]

[a] Anhui Province Key Laboratory of Condensed Matter Physics at Extreme Conditions, High Magnetic Field Laboratory, Chinese Academy of Sciences, Hefei 230031, Anhui, China
[b] University of Science and Technology of China, Hefei 230031, Anhui, China
[c] School of Physical Science and Technology, Southwest University, Chongqing 400715, China
[d] Key Laboratory of Structure and Functional Regulation of Hybrid Materials (Anhui University), Ministry of Education, Institutes of Physical Science and Information Technology, Anhui University, Hefei 230601, China
[e] Collaborative Innovation Center of Advanced Microstructures, Nanjing University, Nanjing 210093, China

ARTICLE INFO

Communicated by K. Nakajima

Keywords:
Flux method
Semiconductor
Two-dimensional materials

ABSTRACT

Gallium selenide (GaSe) shows potential applications in photoelectric fields. Herein, we successfully synthesized GaSe single crystal with large size via improved Bi-flux method. X-ray diffraction pattern and energy dispersive spectrum show the high crystallinity and high homogeneity of GaSe crystal. Optical absorption and photoluminescence measurements illustrate the band gap of the crystal is ~ 2 eV. Photocurrent behavior demonstrates the photoelectric response time is determined to be 2.08 s. The time dependent Raman spectrum shows that GaSe grown by Bi-flux method degenerates slowly under continuous air-exposure. This effective new growth method would be helpful for photoelectric device and two-dimensional materials study based on GaSe.

1. Introduction

Since the discovery of graphene [1], van der Waals coupled layered material has received extensive attention, because it can easily be exfoliated into single or few layers with very high quality and thus opens the possibility to fabricate functional devices not only with ultrathin thickness but also with high performance [2–4]. Two of the most famous examples are MoS_2 and black phosphorus, which show excellent field effect characteristics in single/few layers as have been demonstrated recently [5,6]. GaSe is a layered *p*-type semiconductor with a band gap of about 2 eV [7–9], which is widely used in the field of opto-electronics, nonlinear optics, and terahertz experiments [10–14]. The crystal structure of GaSe is shown in Fig. 1, where each GaSe layer consists of four covalently bonded Se-Ga-Ga-Se atoms forming a hexagonal sheet. These GaSe layers are stacked vertically with or without lateral displacement and coupled to each other by a weak van der Waals interaction. Like other van der Waals coupled layered materials, GaSe can also be easily cleaved into nano-sheets. Recently, good photoresponse and on/off ratio have been realized in the mechanically exfoliated GaSe nanosheet based transistors [7,15], which highlights the application of GaSe. Furthermore, it can potentially combine with other two-dimensional materials to fabricate ultrathin multifunctional devices, such as artificial-heterojunction [2–4], ultrafast photodetector [16], etc. Therefore, besides the direct synthesis of single/few layer GaSe by deposition methods, large GaSe single crystal of high crystalline quality is required urgently to promote further study.

However, two barriers stand in the front of the intensive researches of GaSe. Firstly, the growth of high quality GaSe single crystal is still a challenge. The reasons are chiefly as follows. Vertical Bridgman–-Stockbarger method is commonly used to grow GaSe crystals [17–19], but it requires very accurate thermal conditions and preventing high Se pressure during the growth. Liquid-phase epitaxy is another method that can successfully grow single crystal GaSe, whereas the growth rate is rather small, with the size of the crystal about 4 mm × 7 mm × 300 μm (thickness) under a growth time of 15–30 days [20]. Solid state method [21], and vapor growth method [10] have also been used to synthesis GaSe, while the quality of the crystallization is very poor. Secondly, previous researches manifest the surface of GaSe will oxidize at ambient air conditions, especially for ultrathin nano-flakes cleaved from bulk [22–26]. Moreover, complex conditions, for example oxygen, humidity, and even above-gap laser illumination, can influence the degeneration of GaSe [23]. Hence, exploring one method to synthesize high quality GaSe single crystal and avoid the degeneration of GaSe at ambient air condition can promote further researches of high performance optical-

* Corresponding authors at: Anhui Province Key Laboratory of Condensed Matter Physics at Extreme Conditions, High Magnetic Field Laboratory, Chinese Academy of Sciences, Hefei 230031, Anhui, China (M.L. Tian).
E-mail addresses: xdzhu@hmfl.ac.cn (X. Zhu), tianml@hmfl.ac.cn (M. Tian).

https://doi.org/10.1016/j.jcrysgro.2021.126088
Received 26 November 2020; Received in revised form 22 January 2021; Accepted 18 February 2021
Available online 23 February 2021
0022-0248/© 2021 Elsevier B.V. All rights reserved.

electronic devices.

Here, we report a handy growth-method, using bismuth as flux, to synthesize high quality GaSe single crystal. During the growth process, we vibrate the cooling temperature profile to decrease the number of crystal nuclei. Finally, large GaSe crystal with a typical size of about 15 mm × 5 mm × 500 μm (thickness) is successfully obtained under a growth time of ~ 8 days. X-ray diffractometer (XRD) and energy dispersive spectrometer (EDS) experiments show the as grown crystal is of excellent crystalline quality and high homogeneity. The band gap is studied by absorption and PL experiments, and surface oxidization of GaSe is discussed based on the time dependent Raman spectrum.

2. Experiments

2.1. Growth method

The GaSe single crystals were synthesized via flux method using bismuth (Bi) as flux. The mixed elements of Ga (ingots, 4N5 purity):Se (shots, 4N purity):Bi (ingots, 5N purity) in a molar ratio of 1:1:4 were sealed in a vacuum quartz tube firstly, and then transferred into a box furnace. The furnace was heated to 900 °C and kept at this temperature for 12 h. After that, the furnace was cooled down to 450 °C slowly at a rate of 3 °C/h. At this temperature, Bi flux was removed in a centrifuge out of the furnace. As is shown in the inset of Fig. 2(a), single crystal with a typical size of 4 mm × 4 mm × 400 μm (thickness) can be obtained. We tried slower cooling rate, and found out that it cannot significantly increase the size of the crystal. Then we tried a vibrating cooling temperature profile which is depicted in Fig. 2. Significantly larger crystal with a typical size of ~ 1.5 cm × 5 mm × 500 μm (thickness) can be obtained [see in the inset of Fig. 2(a)]. Too many crystal-nuclei were formed is the reason why small crystals were obtained under common cooling temperature profile, but nucleus number was effectively minimized by vibrating cooling temperature profile. As is shown in Fig. 1, the surface of crystal is covered with equilateral triangles, which corresponds well to the hexagonal in-plane lattice. What's more, the as grown GaSe single crystal can easily be mechanically exfoliated by scotch-tap based method.

2.2. Characterization techniques

All measurements were performed at ambient air condition. The crystal structure of the as grown crystal was analyzed using X-ray diffraction (XRD) with Cu Kα radiation upon a Japan Rigaku TTR-III. Chemical element stoichiometry was analyzed by Energy dispersive spectrum (EDS) equipped on Helios nanoLab 600i. The parameters of semiconductor were measured at room temperature with Van der Pauw method upon a Lake Shore Cryotronics 8400 Series. The optical absorption spectrum was performed on a Cary 5E UV–VisNir Spectrophotometer at room temperature. Photoluminescence (PL) and Raman spectra of the single crystal with fresh surface were measured at room temperature using laser with wavelength of 532 nm. Photocurrent experiment was performed by two-probe method at room temperature, i.e., two gold wires were adhered to the both ends of GaSe crystal to apply voltage and measure current when a 532 nm exciting laser illuminate the surface of GaSe crystal.

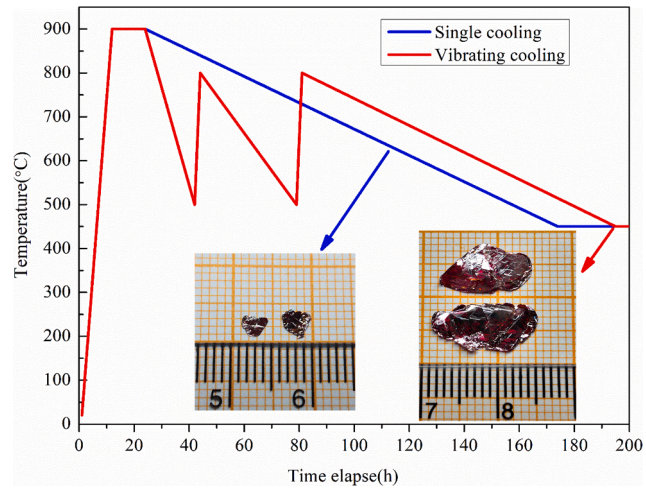

Fig. 2. Single cooling and vibrating cooling temperature profiles of furnace for GaSe single crystal growth. Inset: the photographs of the as-grown single crystals obtained from these two temperature profiles.

3. Results and disscussions

3.1. XRD and EDS

Fig. 3(a) shows the EDS spectrum of the GaSe single crystal. Only Ga and Se elements are detected, indicating the crystal is almost impurity free. The chemical composition ratio of Ga:Se is determined to be 49.8:50.2, which is very close to the ideal chemical stoichiometry of the single crystal. The inset of Fig. 3(a) shows the element mapping of the Ga and Se. Obviously, these two elements are distributed evenly in the size of several tens of microns, indicating a high homogeneity of the single crystal.

Fig. 3(b) shows the XRD pattern of the GaSe single crystal. Six diffraction peaks can be detected in the scan range of 1–80 degrees, which correspond to the crystal plane index of (00 *l*). The deduced lattice parameter *c*, about 1.59 nm, is consistent with previous report [19]. The

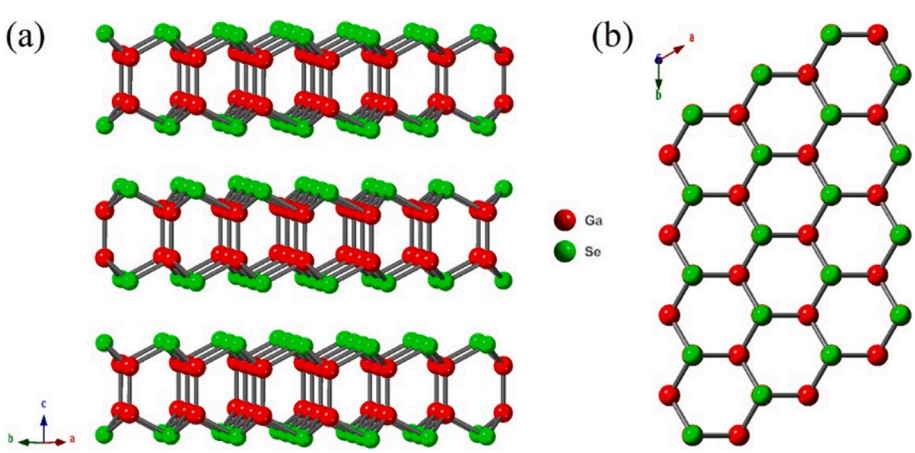

Fig. 1. Structure schematic diagram of layered GaSe. (a) Side view, (b) top view.

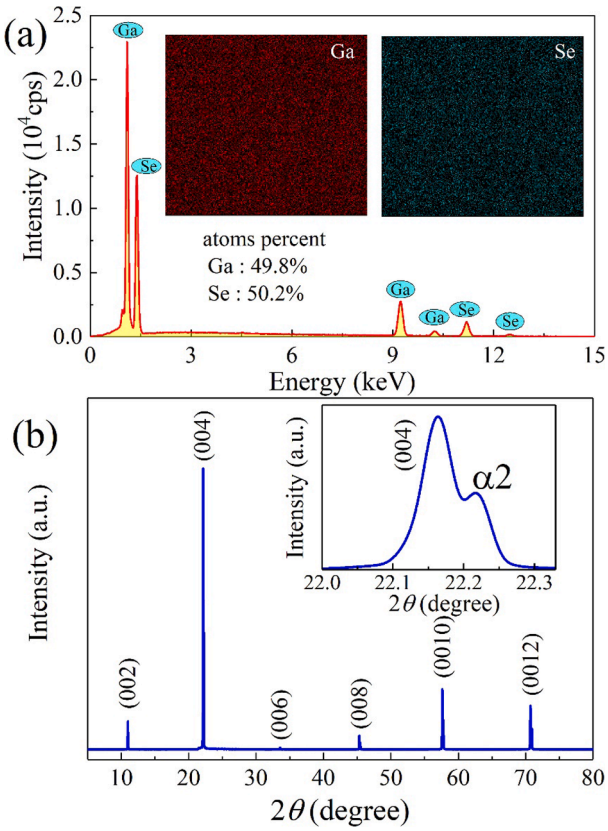

Fig. 3. (a) EDS spectrum of GaSe. Inset: the photogaph of the GaSe crystal. Inset: chemical element mapping of Ga and Se respectively; (b) XRD spectrum of GaSe single crystal. Inset shows the most intense (004) peak. The shoulder located at the right side originates from the Cu Kα_2 radiation.

fact that the full width at the half maximum of the most intensive peak at ~ 22.16 degrees is less than 0.03 degree [inset of Fig. 3(b)], accompanied by an emergence of the diffraction peak due to Cu Kα_2, indicating a very high crystallinity of the crystal. This is further confirmed by the optical photograph as shown in the inset of Fig. 2. One can clearly see that the surface of crystal is covered with equilateral triangles, corresponding well to the hexagonal in-plane lattice as shown in Fig. 1. The triangle-shape formed on crystal surface is consistent with the triangle nanosheets grown by vapor phase deposition method [9,27–30].

3.2. Semiconducting parameters

The basic transport properties of the as grown GaSe measured by Van der Pauw method are listed in Table 1. The carrier type is *p*-type with the carrier concentration of 1.15×10^{16} cm^{-3}, and the Hall mobility is 29.5 cm^2/V·s. The mobility of the as-grown crystal is of the same order compared with that of crystals synthesized by Bridgeman-Stockbarger technique [31]. We notice that the carrier concentration of GaSe grown by Bi-flux method is of an order higher than that of GaSe crystal reported in [31]. However, the starting materials used in reference [31] are 6N which are much higher than ours. Therefore, the higher carrier concentration of GaSe crystal grown by Bi-flux method should originate from the relatively low-purity starting materials.

Table 1
The semiconducting parameters of the as grown GaSe crystal.

parameters		value	unit
Carrier type	–	p-type	–
Carrier concentration	n	1.15×10^{16}	cm^{-3}
Hall mobility	μ_H	29.5	cm^2/V·s
Band gap	E_g	2	eV

3.3. Absorption and photoluminescence spectra

Fig. 4(a) shows the absorption spectrum of the GaSe single crystal measured at room temperature. Only one absorption edge is detected near 2 eV. It is acknowledged that GaSe is an indirect band gap semiconductor [7–9]. However, the width of direct band gap is only 25 meV larger than its indirect one [7–9], which is within the scale of thermal active energy at room temperature. Thus, we roughly use the direct optical transition of $(\alpha E)^2 \propto (E\text{-}E_g)$ to extract the band gap from the absorption edge [32], where α is absorption coefficient, E is the energy of the incident light, and E_g is the band gap of the semiconductor. The optical energy dependence of $(\alpha E)^2$ is plotted in the inset of Fig. 4(a). By extrapolating the absorption edge to the horizontal axis, we obtain the band gap, E_g, of the crystal is about 1.96 eV, which is in line with that reported in previous works [7–9].

Fig. 4(b) shows the photoluminescence (PL) spectrum of the GaSe crystal. Only one strong intensity peak at 2 eV is found near the absorption edge, which corresponds to the optical transition of the direct band gap. The PL spectrum shows the same behavior with that in the previous reports [7,8]. Moreover, the full width at the half maximum of the PL peak is only 0.036 eV, also indicating the high crystal quality. The band gap, 2 eV, determined from the PL spectrum, is about 0.04 eV larger than that determined from the absorption, as shown in the inset of Fig. 4(a), which can be reasonably attributed to the fact that though the width of direct band gap and the indirect gap of GaSe are very close, GaSe is indeed an indirect semiconductor, hence band gap extracted

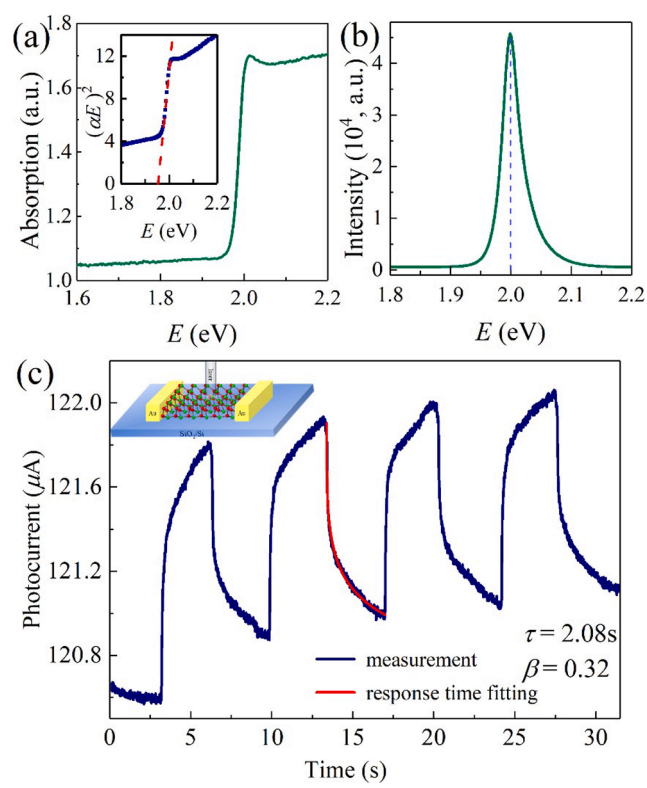

Fig. 4. (a) and (b) show, respectively, the room temperature optical absorption and photoluminescence spectra of the GaSe crystal. Inset in (a): $(\alpha E)^2$ as a function of the photon energy E for the GaSe crystal. The dashed line shows the linear extrapolation for the determination of the band gap. (c) Photoresponse behavior of bulk GaSe applied 0.5 V source-drain voltage, incident light power intensity: 9.4 mW cm^{-2}; inset: schematic representation of the GaSe photocurrent detector; red curve: exponential fitting of response time τ. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

from the direct optical transition might have large error.

3.4. Photocurrent experiment

Fig. 4(c) shows the photoelectric-response behavior of the GaSe bulk single crystal under a driven voltage of 0.5 V with exciting laser of power intensity of 9.4 mW cm^{-2}. By intentionally switching on/off the incident light, a photocurrent of ~ 1 μA is obtained via illuminated current minus dark current. It is noted that the photocurrent changes sharply when the incident light is switched on/off, then followed by a continuous increases/decays. This can be attributed to a persistent photoconductivity effect (PPC) due to the trapping/detrapping process of the carries at deep levels in the band gap of the materials [33]. The photo-response time constant of the crystal, defined as the time interval corresponding to 50% magnitude rise/decay change upon incident light on/off, is determined to be 2.08 s by using the expression,

$$I(t) = I_0 e^{-(t/\tau)^{\beta}} \quad (1)$$

where I_0 is the saturation photocurrent; τ is response time; β is a constant within (0,1) [34,35].

3.5. Air stability of Raman spectrum

To the best of our knowledge, the surface of GaSe will be easily oxidized at ambient air conditions, especially for ultrathin nanosheet cleaved from bulk single crystal [22–26]. Hence, we also determined the degenerate effect of the surface of the as grown GaSe crystal through Raman experiments. Fig. 5 shows the evolution of Raman spectrum along with the time, where Raman spectrum has three main peaks, located at 134.1 cm^{-1}, 215.1 cm^{-1} and 309.9 cm^{-1}, belonging to A^1_{1g}, E^1_{2g}, and A^2_{1g} modes respectively, which is consistent with previous reports [22,25].

The Raman spectra measured on the same sample both for just after cleaving and for at least three weeks air-exposure show the same behavior, while the height of the Raman peaks decreased distinctly with the crtsyal sustainably exposing to the air, implying the continuous degradation of GaSe. Moreover, a broad new peak at ~ 250 cm^{-1} was detected on the sample after exposing to the air for 8 months, which is caused by the formation of α-Se originating from the surface oxidization [22–26].

Although high quality GaSe single crystal was successfully synthesized by Bi-flux method, the surface of GaSe, exposing to the air, degenerates inevitably. Maybe the oxidization is an intrinsic property of GaSe so that the problem of air-unstability cannot be settled by changing growth method. A protection transparent layer should be covered on GaSe for the device design in the future.

4. Conclusion

We successfully grow high quality GaSe single crystal via the improved Bi flux method. EDS and XRD results show that the as grown crystal is of accurate stoichiometry, high homogeneity, and high crystallinity. The electrical properties confirm that the GaSe single crystal is *p*-type semiconductor with carrier concentration of 1.15 × 10^{16} cm^{-3} and Hall mobility of 29.5 cm^2/V·s. The bandgap of the crystal is determined to be 2 eV by the absorption and PL spectra. While the time dependent Raman experiments prove that the as grown GaSe single crystal would oxidize under continuous air-exposure. We hope our crystal growth method of the bulk GaSe can promote further study of this two-dimensional semiconductor GaSe.

CRediT authorship contribution statement

Weiwei Chu: . **Jiyong Yang:** . **Liang Li:** . **Xiangde Zhu:** . **Mingliang Tian:** Supervision.

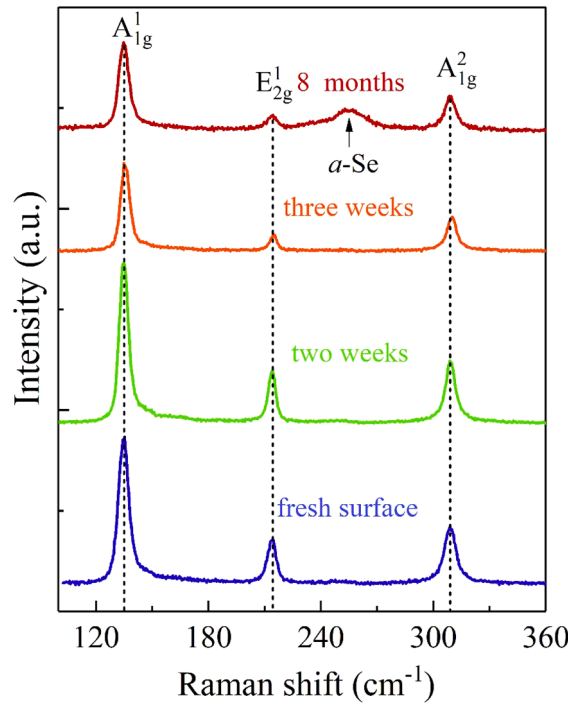

Fig. 5. Time dependent Raman spectrum of GaSe grown by Bi-flux method.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgements

We thank Zhaoming Wang for the help on optical absorption spectrum measurements. We thank Prof. Chong Xiao for the help on semiconducting parameters measurement. Work at High magnetic field lab (Hefei) was supported by Youth Innovation Promotion Association of CAS (Grant No.2017483), National Natural Science Foundation of China (Grants No. U2032163, U19A2093, U1732274), Collaboration Innovation Program of Hefei Science Center, CAS (Grant No. 2019HSC-CIP 001); A portion of this work was supported by the High Magnetic Field Laboratory of Anhui Province.

References

[1] K.S. Novoselov, A.K. Geim, S.V. Morozov, D. Jiang, Y. Zhang, S.V. Dubonos, I. V. Grigorieva, A.A. Firsov, Electric Field Effect in Atomically Thin Carbon Films, Science 306 (2004) 666.
[2] A.K. Geim, I.V. Grigorieva, Van Der Waals Heterostructures, Nature 499 (2013) 419.
[3] H. Zhang, Ultrathin Two-Dimensional Nanomaterials, ACS Nano 9 (2015) 9451.
[4] R. Frisenda, E. Navarro-Moratalla, P. Gant, D. Pérez De Lara, P. Jarillo-Herrero, R. V. Gorbachev, A. Castellanos-Gomez, Recent Progress in the Assembly of Nanodevices and van Der Waals Heterostructures by Deterministic Placement of 2D Materials, Chem. Soc. Rev. 47 (2018) 53.
[5] B. Radisavljevic, A. Radenovic, J. Brivio, V. Giacometti, A. Kis, Single-Layer MoS_2 Transistors, Nat. Nanotechnol. 6 (2011) 147.
[6] L. Li, Y. Yu, G. Ye, Q. Ge, X. Ou, H. Wu, D. Feng, X. Chen, Y. Zhang, Black Phosphorus Field-Effect Transistors, Nat. Nanotechnol. 9 (2014) 372.
[7] P. Hu, Z. Wen, L. Wang, P. Tan, K. Xiao, Synthesis of Few-Layer GaSe Nanosheets for High Performance Photodetectors, ACS Nano 6 (2012) 5988.
[8] C.S. Jung, F. Shojaei, K. Park, J.Y. Oh, H.S. Im, D.M. Jang, J. Park, H.S. Kang, Red-to-Ultraviolet Emission Tuning of Two-Dimensional Gallium Sulfide/Selenide, ACS Nano 9 (2015) 9585.
[9] Z. Ben Aziza, H. Henck, D. Pierucci, M.G. Silly, E. Lhuillier, G. Patriarche, F. Sirotti, M. Eddrief, A. Ouerghi, Van Der Waals Epitaxy of GaSe/Graphene Heterostructure: Electronic and Interfacial Properties, ACS Nano 10 (2016) 9679.

[10] K.R. Allakhverdiev, M.Ö. Yetis, S. Özbek, T.K. Baykara, E. Yu Salaev, Effective Nonlinear GaSe Crystal. Optical Properties and Applications, Laser Phys. 19 (2009) 1092.
[11] L. Leontie, I. Evtodiev, V. Nedeff, M. Stamate, M. Caraman, Photoelectric Properties of Bi_2O_3/GaSe Heterojunctions, Appl. Phys. Lett. 94 (2009) 071903.
[12] S. Lei, L. Ge, Z. Liu, S. Najmaei, G. Shi, G. You, J. Lou, R. Vajtai, P.M. Ajayan, Synthesis and Photoresponse of Large GaSe Atomic Layers, Nano Lett. 13 (2013) 2777.
[13] W. Shi, Y.J. Ding, N. Fernelius, K. Vodopyanov, Efficient, Tunable, and Coherent 018–527-THz Source Based on GaSe Crystal, Opt. Lett. 27 (2002) 1454.
[14] C. Kübler, R. Huber, S. Tübel, A. Leitenstorfer, Ultrabroadband Detection of Multi-Terahertz Field Transients with GaSe Electro-Optic Sensors: Approaching the near Infrared, Appl. Phys. Lett. 85 (2004) 3360.
[15] D.J. Late, B. Liu, J. Luo, A. Yan, H.S.S.R. Matte, M. Grayson, C.N.R. Rao, V. P. Dravid, GaS and GaSe Ultrathin Layer Transistors, Adv. Mater. 24 (2012) 3549.
[16] Z. He, J. Guo, S. Li, Z. Lei, L. Lin, Y. Ke, W. Jie, T. Gong, Y. Lin, T. Cheng, W. Huang, X. Zhang, GaSe/MoS_2 Heterostructure with Ohmic-Contact Electrodes for Fast, Broadband Photoresponse, and Self-Driven Photodetectors, Adv. Mater. Interfaces 7 (2020) 1901848.
[17] N.N. Kolesnikov, E.B. Borisenko, D.N. Borisenko, V.K. Gartman, Influence of Growth Conditions on Microstructure and Properties of GaSe Crystals, J. Cryst. Growth 300 (2007) 294.
[18] M.M. Abdullah, G. Bhagavannarayana, M.A. Wahab, Growth and Characterization of GaSe Single Crystal, J. Cryst. Growth 312 (2010) 1534.
[19] Y. Ni, H. Wu, C. Huang, M. Mao, Z. Wang, X. Cheng, Growth and Quality of Gallium Selenide (GaSe) Crystals, J. Cryst. Growth 381 (2013) 10.
[20] Y. Oyama, T. Tanabe, F. Sato, A. Kenmochi, J. Nishizawa, T. Sasaki, K. Suto, Liquid-Phase Epitaxy of GaSe and Potential Application for Wide Frequency-Tunable Coherent Terahertz-Wave Generation, J. Cryst. Growth 310 (2008) 1923.
[21] M.M. Abdullah, G. Bhagavannarayana, M.A. Wahab, Controlled Synthesis and Structural Characterization of Polycrystalline GaSe, J. Mater. Sci. 45 (2010) 4088.
[22] T.E. Beechem, B.M. Kowalski, M.T. Brumbach, A.E. McDonald, C.D. Spataru, S. W. Howell, T. Ohta, J.A. Pask, N.G. Kalugin, Oxidation of Ultrathin GaSe, Appl. Phys. Lett. 107 (2015) 173103.
[23] A. Bergeron, J. Ibrahim, R. Leonelli, S. Francoeur, Oxidation Dynamics of Ultrathin GaSe Probed through Raman Spectroscopy, Appl. Phys. Lett. 110 (2017) 241901.
[24] M. Rahaman, R.D. Rodriguez, M. Monecke, S.A. Lopez-Rivera, D.R.T. Zahn, GaSe Oxidation in Air: From Bulk to Monolayers, Semicond. Sci. Technol. 32 (2017) 105004.
[25] J. Susoma, J. Lahtinen, M. Kim, J. Riikonen, H. Lipsanen, Crystal Quality of Two-Dimensional Gallium Telluride and Gallium Selenide Using Raman Fingerprint, AIP Adv. 7 (2017) 015014.
[26] Q. Zhao, R. Frisenda, P. Gant, D. Perez de Lara, C. Munuera, M. Garcia-Hernandez, Y. Niu, T. Wang, W. Jie, A. Castellanos-Gomez, Toward Air Stability of Thin GaSe Devices: Avoiding Environmental and Laser-Induced Degradation by Encapsulation, Adv. Funct. Mater. 28 (2018) 1805304.
[27] Y. Zhou, Y. Nie, Y. Liu, K. Yan, J. Hong, C. Jin, Y. Zhou, J. Yin, Z. Liu, H. Peng, Epitaxy and Photoresponse of Two-Dimensional GaSe Crystals on Flexible Transparent Mica Sheets, ACS Nano 8 (2014) 1485.
[28] X. Li, M.-W. Lin, A.A. Puretzky, J.C. Idrobo, C. Ma, M. Chi, M. Yoon, C.M. Rouleau, I.I. Kravchenko, D.B. Geohegan, K. Xiao, Controlled Vapor Phase Growth of Single Crystalline, Two-Dimensional GaSe Crystals with High Photoresponse, Sci. Rep. 4 (2014) 5497.
[29] T. Afaneh, A. Fryer, Y. Xin, R.H. Hyde, N. Kapuruge, H.R. Gutiérrez, Large-Area Growth and Stability of Monolayer Gallium Monochalcogenides for Optoelectronic Devices, ACS Appl. Nano Mater. 3 (2020) 7879.
[30] L. Tan, Q. Liu, Y. Ding, X. Lin, W. Hu, M.-Q. Cai, H. Zhou, Effective Shape-Controlled Synthesis of Gallium Selenide Nanosheets by Vapor Phase Deposition, Nano Res. 13 (2020) 557.
[31] V.G. Voevodin, O.V. Voevodina, S.A. Bereznaya, Z.V. Korotchenko, A.N. Morozov, S.Y. Sarkisov, N.C. Fernelius, J.T. Goldstein, Large Single Crystals of Gallium Selenide: Growing, Doping by In and Characterization, Opt. Mater. 26 (2004) 495.
[32] W. Cai, Y. Zhang, J. Jia, L. Zhang, Semiconducting Optical Properties of Silver/Silica Mesoporous Composite, Appl. Phys. Lett. 73 (1998) 2709.
[33] X. Li, M.B. Jordan, T. Ayari, S. Sundaram, Y. El Gmili, S. Alam, M. Alam, G. Patriarche, P.L. Voss, J. Paul Salvestrini, A. Ougazzaden, Flexible Metal-Semiconductor-Metal Device Prototype on Wafer-Scale Thick Boron Nitride Layers Grown by MOVPE, Sci. Rep. 7 (2017) 786.
[34] S. Cai, G. Parish, J.M. Dell, B.D. Nener, Contribution of Hole Trap to Persistent Photoconductivity in n -Type GaN, J. Appl. Phys. 96 (2004) 1019.
[35] M. Martens, J. Schlegel, P. Vogt, F. Brunner, R. Lossy, J. Würfl, M. Weyers, M. Kneissl, High Gain Ultraviolet Photodetectors Based on AlGaN/GaN Heterostructures for Optical Switching, Appl. Phys. Lett. 98 (2011) 211114.